THE INNER-TRANSLATE CANDIDATE FOR THE BACKWARD-SHIFT NORM

ØISTEIN SØVIK

This proof note has evidence level Sketch: it has not appeared on arXiv or been peer-reviewed and should be checked by the reader. AI assistance was used in adapting and checking the argument and exposition.

> the extremal functions are determined by inner functions with a prescribed value at the origin

This implicit candidate is suggested by Theorem 1.1 and the general norm discussion in Section 1, p. 2, of Konstantinos Bampouras and Adrián Llinares, *The norm of the backward shift on H^4 is $\sqrt[4]{\varphi}$*, arXiv:2605.10469v1. For $2 < p < \infty$, the candidate is the supremum of the quotients obtained from $I - c$, where I is inner, $I(0) = a \in (0,1)$, and $c \in \mathbb{C}$. Here $Bf(z) = (f(z) - f(0))/z$.

Answered: no. Evidence: Sketch.

The candidate is false at $p = 6$: $\Lambda(6) < \|B\|_{H^6 \to H^6}$.

Proof. We first show that the restricted quotient attains its maximum. If it attained the full operator norm, its first-variation equation would force incompatible pole cancellations. Write normalized arc measure as $\mathrm{d}m$ and let P_a be harmonic measure based at $a \in (0,1)$. The boundary values of every inner function with $I(0) = a$ have distribution P_a: for powers this follows from $\int I^n \, \mathrm{d}m = a^n$, and trigonometric polynomial approximation gives the claim. Thus the candidate at $p = 6$ is

$$\Lambda(6)^6 = \sup_{0<a<1, c\in\mathbb{C}} \frac{N(a)}{D_a(c)}, \quad N(a) = \int_{\mathbb{T}} |\zeta - a|^6 \, \mathrm{d}P_a(\zeta), \quad D_a(c) = \int_{\mathbb{T}} |\zeta - c|^6 \, \mathrm{d}P_a(\zeta).$$

For fixed a, strict convexity and conjugation symmetry give a unique real minimizer c_a of D_a. Differentiation shows $D_a'(0) < 0 < D_a'(a)$, hence $0 < c_a < a$.

The maximum over a is attained in $(0,1)$. To see this, the standard circle integral gives

$$N(a) = (1 - a^2)(1 + 4a^2 + a^4). \tag{1}$$

As $a \to 0$, uniqueness of the minimizer gives $c_a \to 0$, so the quotient tends to one. As $a \to 1$, the triangle inequality and $D_a(c_a) \le N(a)$ give $|c_a - a| \le 2N(a)^{1/6} \to 0$. For any sequence $a \to 1$, Fatou's lemma applied to

$$\frac{D_a(c_a)}{1 - a} = \int_{\mathbb{T}} \frac{(1 + a)|\zeta - c_a|^6}{|\zeta - a|^2} \, \mathrm{d}m(\zeta)$$

gives a lower limit at least $2 \int_{\mathbb{T}} |\zeta - 1|^4 \, \mathrm{d}m = 12$. Equation (1) gives $N(a)/(1 - a) \to 12$, hence $N(a)/D_a(c_a) \to 1$. Finally $N(1/2) = 99/64 > 1 = D_{1/2}(0)$, so the maximal quotient exceeds one and is attained at some $a \in (0,1)$, with $c = c_a \in (0, a)$.

Suppose $\Lambda(6) = \|B\|_{H^6 \to H^6}$. Choose the one-zero Blaschke factor $I(z) = (a + z)/(1 + az)$ and put

$$A = 1 - a^2, \quad K = 1 - ac, \quad r = (c - a)/(1 - ac) \in (-a, 0),$$
$$h = I - a = \frac{Az}{1 + az}, \qquad f = I - c = \frac{K(z - r)}{1 + az}.$$

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Then $Bf = h/z$ and $\|Bf\|_6 = \|h\|_6$; thus f would be extremal. Set $\lambda = \|h\|_6^6/\|f\|_6^6$. Varying f by arbitrary multiples of z^n, $n \geq 1$, in the extremal quotient shows that every strictly positive Fourier coefficient of $|h|^4h - \lambda|f|^4f$ vanishes. On $\mathbb{T}$ this function is the rational function

$$Q(z) = \frac{A^5z^3 - \lambda K^5(z-r)^3(1-rz)^2}{(1+az)^3(z+a)^2}.$$

Its only possible pole outside the closed disc is at $z_0 = -1/a$. The positive Fourier coefficients are precisely those of its exterior-pole principal part, so their vanishing forces the numerator to have a zero of order at least three at z_0. Both numerator terms are nonzero there, since $-a < r < 0$. The first and second logarithmic derivatives of $z^3/((z-r)^3(1-rz)^2)$ must therefore vanish at z_0. The first gives

$$-3a + \frac{3a}{1+ar} + \frac{2ar}{a+r} = 0, \qquad 3a^2 + ar = 2$$

because $r \neq 0$. Substitution into the second gives

$$0 = -3a^2 + \frac{3a^2}{(1+ar)^2} + \frac{2a^2r^2}{(a+r)^2} = \frac{a^2(9a^4-4)}{6(1-a^2)^2}.$$

Hence $a^2 = 2/3$, and the first equation then gives $r = 0$, a contradiction. Thus the attained inner-translate quotient is strictly below the operator norm. □